



P12 LOGIC 110nm 1.5V_3.3V Pcell Reference Manual

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This Document: Process Design Kits Reference Manual

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1 Overview

1.1 Introduction

Pcell stands for Process Design Kit. Pcell is a complete set of building blocks that are needed for any full custom integrated circuit design and layout. Pcell Supports the whole full custom IC design flow from schematic and layout creations to post-layout simulations. Elements of Pcell library are based on foundry's process and data that enables user's design with high efficiency and good quality. All verified EDA views enable designer to focus on circuit design in the Cadence Design Environment. The Process Design Kit (Pcell) provides the designer with a basic Cadence Design Framework II infrastructure to start a design.

The Pcell provides the following components:

- Symbols contain the primitives such as mosfet, resistor, capacitor and etc. These include process specific CDF & Callback for the Cadence Composer Schematic Editor.
- auCdl provide CDL netlist capability for Calibre LVS usage.
- auLvs provide function for Assura LVS usage.
- hspiceD provide netlisting capability for Hspice model simulation.
- Technology File & Display Resource File are the heart of a Cadence library. The Technology File defines all layers, layer properties, symbolics and tool information used by the Cadence tools. The Display Resource File defines the display and plotting information such as colors, stipple patterns, etc.
- Pcells are parameterized layout cells for primitive devices such as mosfets, resistors, etc. These Pcells are used within the Cadence Virtuoso Layout Editor to automate custom layout. These are fully compatible with the automated layout tools such as Virtuoso-XL and Virtuoso Custom Router.

The purpose of this Reference Manual is to describe the technical information of the P12 LOGIC 110nm 1.5V_3.3V 1P5M Process Design Kit ("Pcell"). The Pcell is a device library that supports the whole IC design flow from schematic creations to post-layout simulations. This Pcell was tested on Cadence IC 6.1.6 design platform.

1.2 Foundry Data

The set of Pcell (Pcell, netlisting, simulation views, etc) and involved technology files refer to PSMC process data. Below Foundry Process Data lists all the technical publications by record. The Pcell supports LOGIC 0.11um SC 1.5V_3.3V process:

Table 1-1: Foundry Data List

File Name	Version
P12 LOGIC 110nm 1.5V_3.3V Design Service SPICE Model 1.5V	
P12 LOGIC 110nm 1.5V_3.3V Design Service SPICE Model 3.3V	
P12 LOGIC 110nm 1.5V_3.3V Design Document DRC Command File	0.10
P12 LOGIC 110nm 1.5V_3.3V Design Document LVS Command File	0.54
P12_LOGIC_110nm_1.5V_3.3V_Design_Rules	0.57

1.3 BEOL Support

There are four BEOL top metal processes provided in PSMC LOGIC 0.11um SC 1.5V_3.3V technology. The Pcell support both BEOL options. The options are: 1P5M and 1P4M.

2 Supported Devices

2.1 Mos Devices

Table 2-1: Mos Devices

Description	name	symbol	auCdl	auLvs	layout	hspiceD	ivPcell	cdf	callback	VXL
1.5V NMOS	lvn	x	x	x	x	x	x	x	x	x
1.5V PMOS	lvp	x	x	x	x	x	x	x	x	x
3.3V NMOS	hvn	x	x	x	x	x	x	x	x	x
3.3V PMOS	hvp	x	x	x	x	x	x	x	x	x

2.2 Resistors

Table 2-2: Resistors

Description	name	symbol	auCdl	auLvs	layout	hspiceD	ivPcell	cdf	callback	VXL
N+Poly	rnpoly	x	x	x	x	x	x	x	x	x
P+Poly	rppoly200	x	x	x	x	x	x	x	x	x
N+Poly	rnpolync	x	x	x	x	x	x	x	x	x
P+Poly	rppolync	x	x	x	x	x	x	x	x	x
	rnaa	x	x	x		x	x	x	x	x
	rpaa	x	x	x		x	x	x	x	x
	rpp	x	x	x		x	x	x	x	x
	rnp	x	x	x		x	x	x	x	x

2.3 Diodes

Table 2-3: Diodes

Description	name	symbol	auCdl	auLvs	layout	hspiceD	ivPcell	cdf	callback	VXL
P+/Nwell	ppnw	x	x	x		x	x	x	x	x
N+/Pwell	nppw	x	x	x		x	x	x	x	x

2.4 Capacitors

Table 2-4: Capacitors

Description	name	symbol	auCdl	auLvs	layout	hspiceD	ivPcell	cdf	callback	VXL
Pwell+3S (Dep. NMOS)	c3s	x	x	x	x	x	x	x	x	x
	cmim1p5f	x	x	x	x	x	x	x	x	x

2.5 Bipolars

Table 2-5: Bipolars

Description	name	symbol	auCdl	auLvs	layout	hspiceD	ivPcell	cdf	callback	VXL
PNP	pnp	x	x	x	x	x	x	x	x	x
NPN	nnp	x	x	x	x	x	x	x	x	x

3 Symbol Display Information

3.1 Mos Devices

Figure 3-1: NMOS Symbol

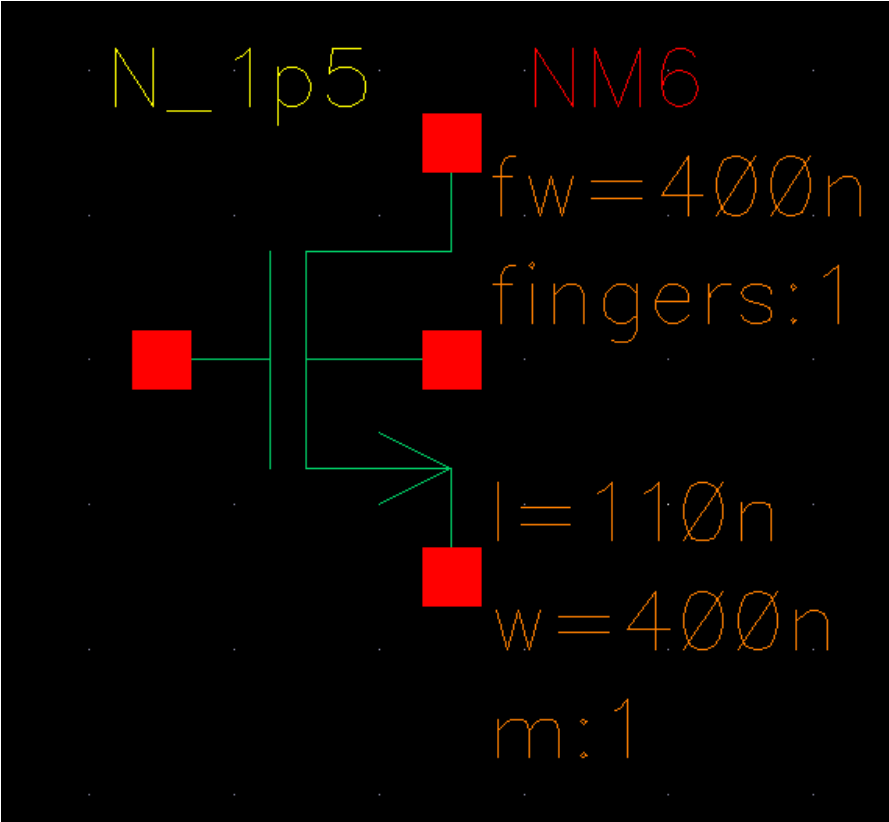


Table 3-1: NMOS Symbol Display

Symbol CDF Name	Description
w	Total Channel Width
l	Channel Length
m	Multiplier

Figure 3-2: PMOS Symbol

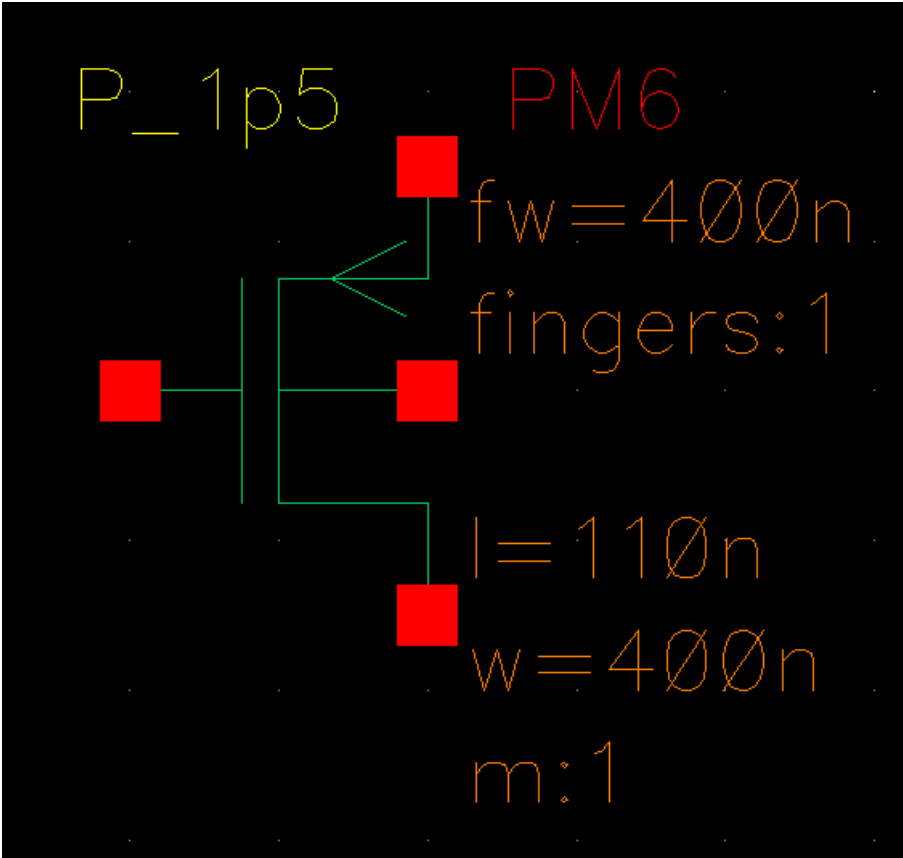


Table 3-2: PMOS Symbol Display

Symbol CDF Name	Description
w	Total Channel Width
l	Channel Length
m	Multiplier

3.2 Resisotr Devices

Figure 3-3: Resistor Symbol

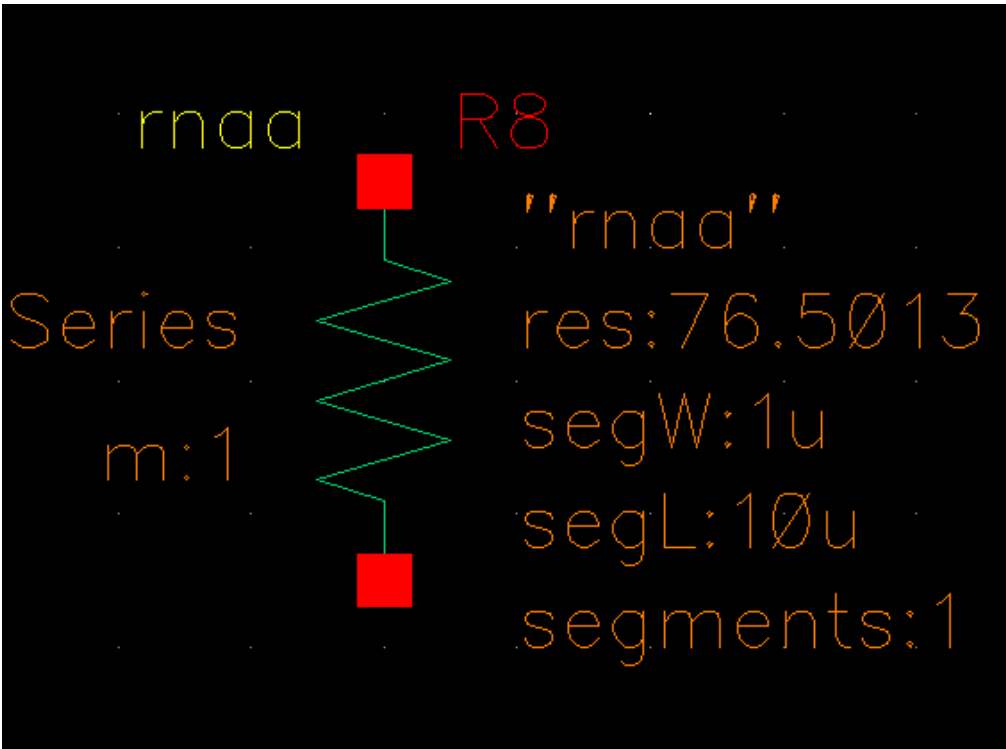


Table 3-3: Resistor Symbol Display

Symbol CDF Name	Description
res	Total Resistance
segW	Segment Width
segL	Segment Length
segments	Number of Segments
Series	Series or Parellel
m	Multiplier

3.3 Diode Devices

Figure 3-4: Diode Symbol

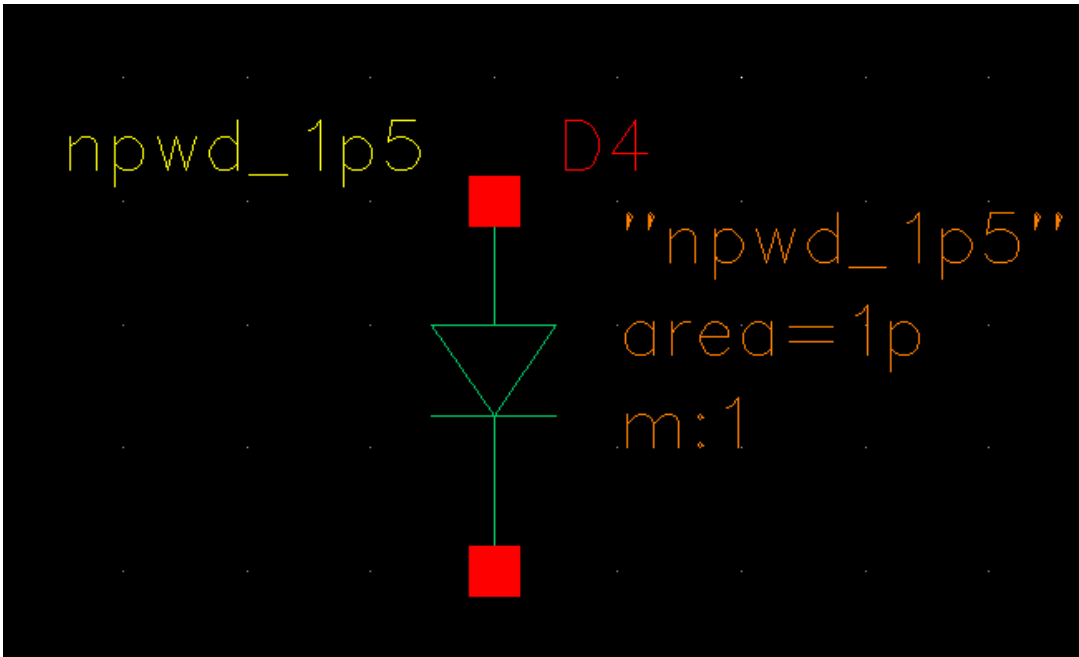


Table 3-4: Diode Symbol Display

Symbol CDF Name	Description
area	Junction Area
m	Multiplier

3.4 Capacitor Devices

Figure 3-5: MIM Symbol

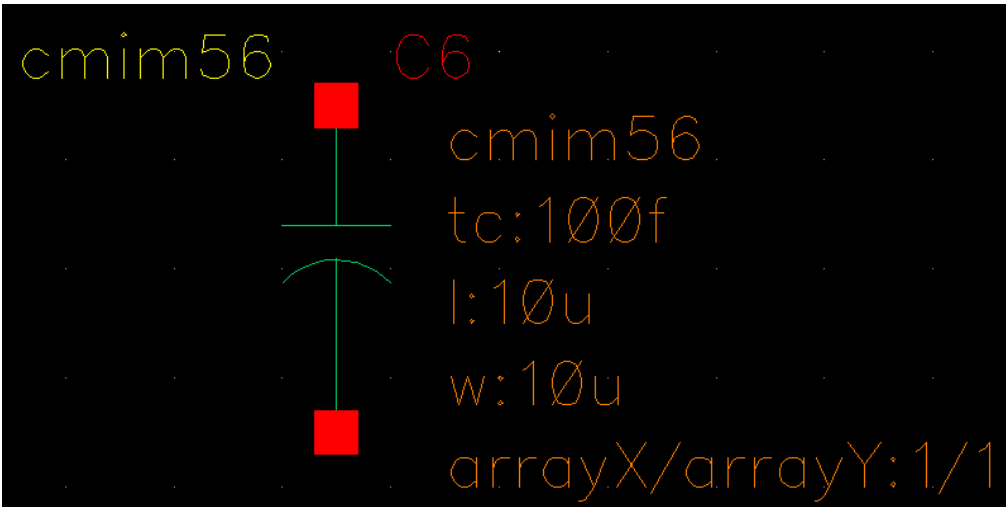


Table 3-5: MIM Symbol Display

Symbol CDF Name	Description
c	Total Capacitance
l	Unit Length
w	Unit Length
arrayX/arrayY	The Array Value

3.5 Bipolar Devices

Figure 3-6: NPN Symbol

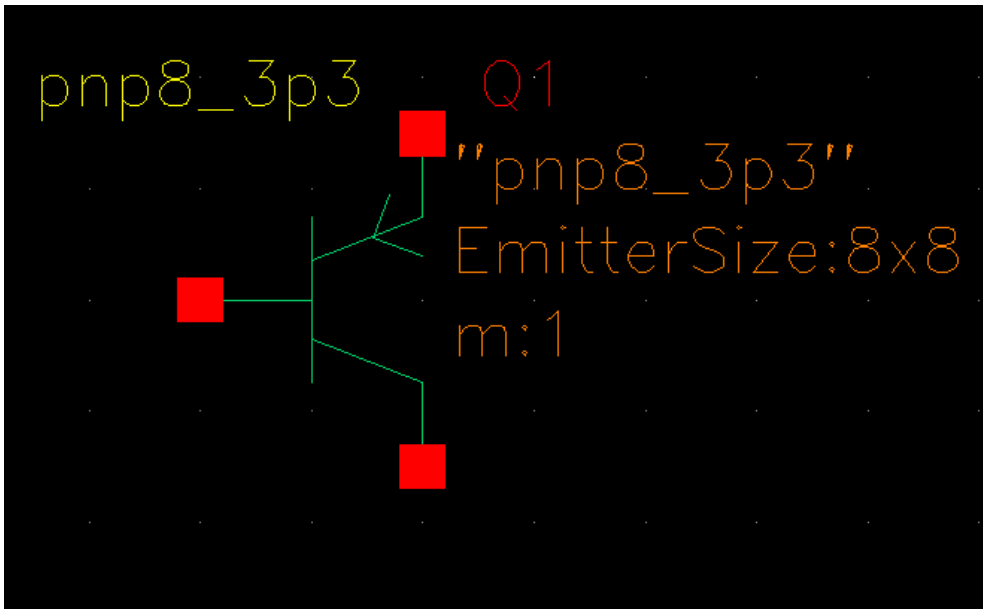


Table 3-6: NPN Symbol Display

Symbol CDF Name	Description
EmitterSize	Emitter Size
m	Multiplier

4 Pcell CDF Parameters

4.1 Mos Devices

Figure 4-1: Mos CDF

CDF Parameter	Value	Display
Model Name	N_1p5	off
Description	1.5V NMOS	off
Multiplier	1	off
Length	110n M	off
Total Width	400n M	off
Finger Width	400n M	off
Fingers	1	off
Apply Threshold	☐	off
Finger Threshold	200u M	off
Gate Connection	☐ None ☐ Top ☐ Bottom ☒ Both	off
S/D Metal Width	240n M	off
Switch S/D	☐	off
Bodytie Type	Detached	off
Left Tap	☒	off
Right Tap	☒	off
Bottom Tap	☒	off
Top Tap	☒	off
Tap Extension	1	off
Tap Contact Rows	1	off

Table 4-1: Mos CDF

Name	Prompt	Description
model	Model Name	The value for this parameter represents the device model name.
description	Description	Display the device description.
m	Multiplier	M factor: the number of devices connected in parallel represented by the instance.
l	Length	Channel length (unit: M).
w	Total Width	Total width of the channel : typically equal to fw*fingers (unit M).
fw	Finger Width	Channel width per finger (unit M).
fingers	Fingers	Number of fingers.
threshold	Threshold	A threshold value for finger width (unit M). Number of fingers is calculated to keep the finger width less than or equal to the threshold value. This parameter is applied only when the 'Apply Threshold' switch is on.
applyThresh	Apply Threshold	Toggle switch for the effective of the 'Threshold' value.

Name	Prompt	Description
connectGates	Gate Connection	Cyclic field with 'None' 'Top' 'Bottom' and 'Both' choices.
connectSD	S/D Connection	Cyclic field with 'None' 'Source' 'Drain' or 'Both' choices. The source metals are connected if the value is 'Source' or 'Both'. The drain metals are connected if the value is 'Drain' or 'Both'.
sdMtlWidth	S/D Metal Width	The width of the metal on the source and drain areas. The source and drain diffusion areas are scaled accordingly to allow larger widths for the metal. Additional columns of contact will fit to the area.
editAreaPerim	Edit Area & Perim	Toggle switch for calculation of Drain/Source Area/Periphery.
ad	Drain Diffusion Area	Drain diffusion area; editable only if 'Edit Area & Perim' switch is turned on.
as	Source Diffusion Area	Source diffusion area; editable only if 'Edit Area & Perim' switch is turned on.
pd	Drain Diffusion Periphery	Drain diffusion periphery; editable only if 'Edit Area & Perim' switch is turned on.
ps	Source Diffusion Periphery	Source diffusion periphery; editable only if 'Edit Area & Perim' switch is turned on.

4.2 Resistor Devices

Figure 4-2: Res CDF

CDF Parameter	Value	Display
Model name	rnaa	off
Description	Rs_N+AA_CO	off
Multiplier	1	off
Calculated Parameter	Resistance	off
Total Resistance(ohms)	76.5013 Ohms	off
Total Width(M)	1u M	off
Total Length(M)	10u M	off
Segment Resistance(ohms)	76.5013 Ohms	off
Segment Width(M)	1u M	off
Segment Length(M)	10u M	off
Number of Segments	1	off
Resistor Connection	☒ Series ☐ Parallel	off
Rs (ohms/square)	10	off
Segment Spacing(M)	280.00n M	off
Contact Rows	1	off
Contact Columns	3	off
Left Dummy	☐	off
Right Dummy	☐	off
Sheet Resistivity(ohms)	8.0965 Ohms	off
Delta Width(M)	-58.3475n M	off
End Resistance(M)	0 M	off
Temperature Coefficient 1	3.150989E-03	off
Temperature Coefficient 2	1.825093E-06	off

Table 4-2: Res CDF

Name	Prompt	Description
model	Model Name	The value for this parameter represents the device model name.
description	Description	Display the device description.
m	Multiplier	M factor: the number of devices connected in parallel represented by the instance.
res	Total Resistance	Device resistance value.
w	Total Width	Total resistance width-in parallel: segments*(segment width)
l	Total Legnth	Total resistance length-in series: segments*(segment length)
calculatedParam	Calculated Parameter	Is used to select whether resistance or length is to be calculated in callbacks.
segments	Segments	Device segments value.
connection	Connection	Cyclic choices of Series or Parallel connection.
segSpacing	Segments Spacing	Spacing between resistance segments.
seriesSeg	Series Segment	Display the value of series segment.
parallelSeg	Parallel Segment	Display the value of parallel segment.
leftDummy	Left Dummy	"Dummy" parameters are used to place a ""dummy"" resistor segment on the left or right side of the resistor segments. This feature is useful in critical applications where resistor matching is required."
rightDummy	Right Dummy	"Dummy" parameters are used to place a ""dummy"" resistor segment on the left or right side of the resistor segments. This feature is useful in critical applications where resistor matching is required."
contOnDummy	Contact On Dummy	The parameter is used to leave the dummy segment floating without contacts or to place contacts for eventual routing.
contRows	Contact Rows	Device Contact Rows Number.
contCols	Contact Columns	Device Contact Columns Number.
rsh	Sheet Resistance	Display the device rsh value.
rend	End Resistance	Display the device rend value.
dw	Delta Width	Display the device dw value.
dl	Delta Length	Display the device dl value
tc1	Temperature coefficient 1	Display the device tc1 value
tc2	Temperature coefficient 2	Display the device tc2 value.

4.3 Diode Devices

Figure 4-3: Diode CDF

CDF Parameter	Value	Display
Model name	npwd_1p5	off ☐
Description	LV LVT N+/Pwell	off ☐
Multiplier	1	off ☐
Width	1u	off ☐
Length	1u	off ☐
Area	1p	off ☐
Periphery of Junction	4u	off ☐

Table 4-3: Diode CDF

Name	Prompt	Description
model	Model name	The value for this parameter represents the device model name. This parameter is non-editable.
description	Description	Display the device description.
m	Multiplier	M factor: the number of devices connected in parallel represented by the instance.
calcParam	Calculate Parameter	"Select which parameter to calculate: ""area"" ""width"" or ""length"". The selected parameter is then made non-editable while the other two are editable."
area	Device area	"The junction area; editable only if ""Calculate
w	Width	"Width of device; editable only if ""Calculate
l	Length	"Length of device; editable only if ""Calculate
pj	Periphery of junction	Periphery of the junction. This parameter is non-editable.
applyDNW	Apply Deep NW	Apply Deep Nwell

4.4 Capacitor Devices

Figure 4-4: MIM CDF

CDF Parameter	Value	Display
Model Name	cmin56	off ☐
Description	MIM typical Capacitor	off ☐
Multiplier	1	off ☐
Calculate Parameter	capacitance ☐	off ☐
Total Capacitance	100f F	off ☐
Unit Capacitance	100f F	off ☐
Width	10u M	off ☐
Length	10u M	off ☐
Array X	1	off ☐
Array Y	1	off ☐
Top Plate Connection	☒	off ☐
Bottom Plate Connection	☒ None ☐ Top ☐ Bottom ☐ Both	off ☐
Capacitor Area	100p	off ☐
Capacitor Perimeter	40u	off ☐
Area Capacitance (F/M^2)	0.001	off ☐
Fringe Capacitance (F/M)	0	off ☐

Table 4-4: MIM CDF

Name	Prompt	Description
model	Model name	The value for this parameter represents the device model name. This parameter is non- editable.
description	Description	Display the device description.
m	Multiplier	M factor: the number of devices connected in parallel represented by the instance.
calcParam	Calculate Parameter	"Select which parameter to calculate: ""capacitance"" ""width"" or ""length"". The selected parameter is then made non-editable while the other two are editable."
totalCap	Total Capacitance	"Total Capacitance"
equal to: cap*(arrayX)*(arrayY)"		
cap	Capacitance	"Device Capacitance Value; editable only if ""Calculated Parameter"" is not ""area""."

Name	Prompt	Description
w	Width	"Width of device; editable only if ""Calculate Parameter"" is not set to ""width""."
l	Length	"Length of device; editable only if ""Calculate Parameter"" is not set to ""length""."
arrayX	Array X	X coordinate Array.
arrayY	Array Y	Y coordinate Array.
ca	Area capacitance (F/M^2)	Capacitance per area. This parameter is non-editable.
cf	Fringe capacitance (F/M)	Capacitance per perimeter. This parameter is non-editable.
topPConn	Top Plate Connection	Top Plate Connection
botPConn	Bottom Plate Connection	Bottom Plate Connection

Figure 4-5: NW Cap CDF

CDF Parameter	Value	Display
Model Name	cmm56	off ☐
Description	MIM typical Capacitor	off ☐
Multiplier	1	off ☐
Calculate Parameter	capacitance ☐	off ☐
Total Capacitance	100f F	off ☐
Unit Capacitance	100f F	off ☐
Width	10u M	off ☐
Length	10u M	off ☐
Array X	1	off ☐
Array Y	1	off ☐
Top Plate Connection	☒	off ☐
Bottom Plate Connection	☒ None ☐ Top ☐ Bottom ☐ Both	off ☐
Capacitor Area	100p	off ☐
Capacitor Perimeter	40u	off ☐
Area Capacitance (F/M^2)	0.001	off ☐
Fringe Capacitance (F/M)	0	off ☐

Table 4-5: NW Cap CDF

Name	Prompt	Description
model	Model name	The value for this parameter represents the device model name. This parameter is non- editable.
description	Description	Display the device description.

Name	Prompt	Description
m	Multiplier	M factor: the number of devices connected in parallel represented by the instance.
calcParam	Calculate Parameter	"Select which parameter to calculate: ""capacitance"" ""width"" or ""length"". The selected parameter is then made non-editable while the other two are editable."
cap	Capacitance	"Device Capacitance Value; editable only if ""Calculated Parameter"" is not ""area""."
w	Gate Width	"Width of device; editable only if ""Calculate Parameter"" is not set to ""width""."
l	Gate Length	"Length of device; editable only if ""Calculate Parameter"" is not set to ""length""."
connectGate	Gate Connection	Gate Connection Choice: None Top Bottom or Both.
ca	Area capacitance (F/M^2)	Capacitance per area. This parameter is non-editable.
cf	Fringe capacitance (F/M)	Capacitance per perimeter. This parameter is non-editable.
applyDNW	Apply Deep NW	Apply Deep Nwell

4.5 Bipolar Devices

Figure 4-6: Bipolar CDF

CDF Parameter	Value	Display
Model name	pnp8_3p3	off ☐
Description	8x8 lateral PNP bipolar	off ☐
Emitter Size	8x8	off ☐
Area	64p	off ☐
Multiplier	1	off ☐
EmitterLength	8u	off ☐
EmitterWidth	8u	off ☐

Table 4-6: Bipolar CDF

Name	Prompt	Description
model	Model Name	The value for this parameter represents the device model name.
description	Description	Display the device description.
m	Multiplier	M factor: the number of devices connected in parallel represented by the instance.
emitterSize	Emitter Size	Cyclic field with '8x8' choices. Select the target value for the BJT dimension.
emitterLength	Emitter Length	Display the BJT emitter area.
emitterWidth	Emitter Width	Display the BJT emitter width.

5 REVISION HISTORY

Version	Issue Date	Author	Summary of Changes
0.00	2023/10/17	PSMC	Initial Creation Only symbol
0.01	2024/01/08	PSMC	Initial Creation
0.02		PSMC	Fix display error
0.03	2024/04/12	PSMC	Add 1p5f mim cap

Appendix-I

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Table 3-3: Resistor Symbol Display 10

Table 3-4: Diode Symbol Display 11

Table 3-5: MIM Symbol Display 12

Table 3-6: NPN Symbol Display..... 12

Table 4-1: Mos CDF 14

Table 4-2: Res CDF 17

Table 4-3: Diode CDF 18

Table 4-4: MIM CDF..... 19

Table 4-5: NW Cap CDF 20

Table 4-6: Bipolar CDF..... 21

Appendix-II

Figure 3-1: NMOS Symbol 8

Figure 3-2: PMOS Symbol 9

Figure 3-3: Resistor Symbol..... 10

Figure 3-4: Diode Symbol..... 11

Figure 3-5: MIM Symbol 12

Figure 3-6: NPN Symbol 12

Figure 4-1: Mos CDF..... 14

Figure 4-2: Res CDF 17

Figure 4-3: Diode CDF..... 18

Figure 4-4: MIM CDF..... 19

Figure 4-4: NW Cap CDF 20

Figure 4-5: Bipolar CDF 21